8.6 Linear Algebra for Statistics and Probability

Statistics deals with data, often in large quantities. Since data tends to go into rectangular matrices, we expect to see A^TA. The least squares problem $A\widehat{x} \approx \boldsymbol{b}$ is ***linear regression.*** The best solution $\widehat{x}$ fits m observations by $n < m$ parameters. This is a fundamental application of linear algebra to statistics.

This section goes beyond $A^TA\widehat{x} = A^T\boldsymbol{b}$. These unweighted equations assume that the measurements $b_1, \ldots, b_m$ are equally reliable. When there is good reason to expect higher accuracy (lower variance) in some b_i, those equations should be weighted more heavily. *With what weights* $w_1, \cdots, w_m$? And if the b_i are not independent, a ***covariance matrix*** Σ gives the statistics of the errors. Here are key topics in this section:

1. Weighted least squares and $A^TCA\widehat{x} = A^TC\boldsymbol{b}$
2. Variances $\sigma_1^2, \ldots, \sigma_m^2$ and the covariance matrix $\boldsymbol{\Sigma}$
3. Important probability distributions: binomial, Poisson, and normal
4. Principal Component Analysis (PCA) to find combinations with greatest variance.

Weighted Least Squares

To include weights in the m equations $A\boldsymbol{x} = \boldsymbol{b}$, multiply each equation i by a weight w_i. Put those m weights into a diagonal matrix W. *We are replacing* $A\boldsymbol{x} = \boldsymbol{b}$ *by* $WA\boldsymbol{x} = W\boldsymbol{b}$. The equations are no more and no less solvable—we expect to use least squares.

The least squares equation $A^TA\widehat{x} = A^T\boldsymbol{b}$ changes to $(WA)^TWA\widehat{x} = (WA)^TW\boldsymbol{b}$. The matrix $C = W^TW$ is inside $(WA)^TWA$, in the middle of weighted least squares.

Weighted least squares $\quad C = W^TW$ *is in the n equations for* $\widehat{x}$ $\quad A^TCA\widehat{x} = A^TC\boldsymbol{b}$ (1)

When $n = 1$ and $A =$ column of 1's, $\widehat{x}$ changes from an average to a weighted average:

$$\textbf{Simplest case} \quad \widehat{x} = \frac{b_1 + \cdots + b_m}{m} \quad \text{changes to} \quad \widehat{x}_W = \frac{w_1^2 b_1 + \cdots + w_m^2 b_m}{w_1^2 + \cdots + w_m^2}. \tag{2}$$

This average $\widehat{x}_W$ gives greatest weight to the observations b_i that have the largest w_i. We always assume that errors have *zero mean.* (Subtract the mean if necessary, so there is no one-sided bias in the measurements.)

How should we choose the weights w_i? This depends on the reliability of b_i. If that observation has variance σ_i^2, then the root mean square error in b_i is σ_i. **When we divide the equations by** $\sigma_1, \ldots \sigma_m$ (left side together with right side), **all variances will equal 1.** So the weight is $w_i = 1/\sigma_i$ and the diagonal of $C = W^TW$ contains the numbers $1/\sigma_i^2$.

The statistically correct matrix is $C = \operatorname{diag}(1/\sigma_1^2, \ldots, 1/\sigma_m^2)$.

This is correct provided the errors e_i and e_j in different equations are statistically independent. If the errors are dependent, off-diagonal entries show up in the covariance matrix $\boldsymbol{\Sigma}$. The good choice is still $C = \boldsymbol{\Sigma}^{-1}$ as described in this section.

Mean and Variance

The two crucial numbers for a random variable are its **mean** m and its **variance** σ^2. The "expected value" E[e] is found from the probabilities $p_1, p_2, \ldots$ of the possible errors $e_1, e_2, \ldots$ (and the variance σ^2 is always measured around the mean).

For a discrete random variable, the error e_j has probability p_j (the p_j add to 1):

$$\textbf{Mean } m = \mathrm{E}[e] = \sum e_j p_j \quad \textbf{Variance } \sigma^2 = \mathrm{E}[(e-m)^2] = \sum (e_j - m)^2 p_j \tag{3}$$

Example 1 Flip a fair coin. The result is 1 (for heads) or 0 (for tails). Those events have equal probabilities $p_0 = p_1 = 1/2$. The mean is $m = 1/2$ and the variance is $\sigma^2 = 1/4$:

$$\text{Mean} = (0)\frac{1}{2} + (1)\frac{1}{2} \qquad \text{Variance} = \left(0 - \frac{1}{2}\right)^2 \frac{1}{2} + \left(1 - \frac{1}{2}\right)^2 \frac{1}{2} = \frac{1}{4}.$$

Example 2 (**Binomial**) Flip the fair coin N times and count heads. With 3 flips, we see $M = 0, 1, 2$, or 3 heads. The chances are $1/8, 3/8, 3/8, 1/8$. There are three ways to see $M = 2$ heads: HHT, HTH, and THH, and only HHH for $M = 3$ heads.

For all N, the number of ways to see M heads is the binomial coefficient "N *choose* M". Divide by the total number 2^N of all possible outcomes to get the probability for each M:

$$\begin{matrix}\textbf{M heads in}\\ \textbf{N coin flips}\end{matrix} \quad p_M = \frac{1}{2^N}\binom{N}{M} = \frac{1}{2^N}\frac{N!}{M!(N-M)!} \qquad \text{Check } \frac{1}{2^3}\frac{3!}{2!\,1!} = \frac{3}{8} \tag{4}$$

Gamblers know this instinctively. The probabilities p_M add to $\left(\frac{1}{2} + \frac{1}{2}\right)^N = 1$. The mean value of the number of heads is $m = N/2$. The variance around m turns out to be $\sigma^2 = N/4$. The standard deviation $\sigma = \sqrt{N}/2$ measures the expected spread around the mean.

Example 3 (**Poisson**) A very unfair coin (small $p << \frac{1}{2}$) is flipped very often (large N). **The product $\lambda = pN$ is kept fixed.** The high probability of tails is $1 - p$ each time. So the chance p_0 of no heads in N flips (tails every time) is $(1-p)^N = (1 - \lambda/N)^N$. For large N this approaches $e^{-\lambda}$. The probability p_j of j heads in N very unfair flips comes out neatly in terms of the crucial number $\lambda = pN$:

$$\textbf{Poisson probabilities} \quad p_j = \frac{\lambda^j}{j!}e^{-\lambda} \quad \textbf{Mean } m = \lambda \quad \textbf{Variance } \sigma^2 = \lambda \tag{5}$$

Poisson applies to counting infrequent events (low p) over a long time T. Then $\lambda = pT$.

A *continuous* random variable will have a probability *density* function $p(x)$ instead of $p_1, p_2, \ldots$. "An outcome between x and $x + dx$ has probability $p(x)\,dx$." The total probability is $\int p(x)\,dx = 1$, since some outcome must happen. Sums become integrals:

$$\textbf{Mean } m = \textbf{Expected value} = \int x p(x)\,dx \quad \textbf{Variance } \sigma^2 = \int (x-m)^2 p(x)\,dx. \tag{6}$$

The outstanding example of a probability density function $p(x)$ (called the pdf) is the ***normal distribution*** $\mathbf{N}(0, \sigma)$. This has mean zero by symmetry. Its variance is σ^2:

$$\textbf{Normal (Gaussian)} \qquad p(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-x^2/2\sigma^2} \quad \text{with} \quad \int_{-\infty}^{\infty} p(x)\,dx = 1. \tag{7}$$

The graph of $p(x)$ is the famous bell-shaped curve. The integral of $p(x)$ from $-\sigma$ to σ is the probability that a random sample is less than one standard deviation σ from the mean. This is near $2/3$. MATLAB's **randn** uses the normal distribution with $\sigma = 1$.

This normal $p(x)$ appears everywhere because of the ***Central Limit Theorem***: The average over many independent trials of another distribution (like binomial) will approach a normal distribution as $N \to \infty$. A shift produces $m = 0$ and rescaling produces $\sigma = 1$.

$$\textbf{Normalized headcount} \qquad x = \frac{M - \text{mean}}{\sigma} = \frac{M - N/2}{\sqrt{N}/2} \longrightarrow \text{Normal } \mathbf{N}(0, 1).$$

The Covariance Matrix

Now run m different experiments at once. They might be independent, or there might be some correlation between them. Each measurement $\boldsymbol{b}$ is now a *vector* with m components. Those components are the outputs b_i from the m experiments.

If we measure distances from the means m_i, each error $e_i = b_i - m_i$ has *mean zero*. If two errors e_i and e_j are *independent* (no relation between them), their product $e_i e_j$ also has mean zero. But if the measurements are by the same observer at nearly the same time, the errors e_i and e_j could tend to have the same sign or the same size. ***The errors in the m experiments could be correlated.*** The products $e_i e_j$ are weighted by p_{ij} (their probability): *covariance* $\sigma_{ij} = \sum\sum p_{ij} e_i e_j$. The sum of $e_i^2 p_{ii}$ is the variance σ_i^2:

$$\textbf{Covariance} \qquad \sigma_{ij} = \sigma_{ji} = \mathbf{E}[e_i e_j] = \textbf{expected value of } (e_i \textbf{ times } e_j). \tag{8}$$

This is the (i, j) and (j, i) entry of the ***covariance matrix*** $\boldsymbol{\Sigma}$. The (i, i) entry is $\sigma_{ii} = \sigma_i^2$.

Example 4 **(Multivariate normal)** For m random variables, the probability density function moves from $p(x)$ to $p(\boldsymbol{b}) = p(b_1, \ldots, b_m)$. The normal distribution with mean zero was controlled by one positive number σ^2. Now $p(\boldsymbol{b})$ is controlled by an m by m positive definite matrix $\boldsymbol{\Sigma}$. This is the covariance matrix and its determinant is $|\boldsymbol{\Sigma}|$:

$$p(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-x^2/2\sigma^2} \quad \text{becomes} \quad p(\boldsymbol{b}) = \frac{1}{(2\pi)^{m/2} |\boldsymbol{\Sigma}|^{1/2}} e^{-\boldsymbol{b}^{\mathrm{T}} \boldsymbol{\Sigma}^{-1} \boldsymbol{b}/2}$$

The integral of $p(\boldsymbol{b})$ over m-dimensional space is 1. The integral of $\boldsymbol{b}\boldsymbol{b}^{\mathrm{T}} p(\boldsymbol{b})$ is $\boldsymbol{\Sigma}$.

The good way to handle that exponent $-\boldsymbol{b}^{\mathrm{T}}\Sigma^{-1}\boldsymbol{b}/2$ is to use the eigenvalues and orthonormal eigenvectors of Σ (*linear algebra enters here*). When $\Sigma = Q\Lambda Q^{\mathrm{T}} = Q\Lambda Q^{-1}$, replacing $\boldsymbol{b}$ by $Q\boldsymbol{c}$ will split $p(\boldsymbol{b})$ into m one-dimensional normal distributions:

$$\exp\left(-\boldsymbol{b}^{\mathrm{T}}\Sigma^{-1}\boldsymbol{b}/2\right) = \exp\left(-\boldsymbol{c}^{\mathrm{T}}\Lambda^{-1}\boldsymbol{c}/2\right) = \left(e^{-c_1^2/2\lambda_1}\right)\cdots\left(e^{-c_m^2/2\lambda_m}\right).$$

The determinant has $|\Sigma|^{1/2} = |\Lambda|^{1/2} = (\lambda_1 \cdots \lambda_m)^{1/2}$. Each integral over $-\infty < c_i < \infty$ is back to one dimension, where $\lambda = \sigma^2$. Notice the wonderful fact that after any linear transformation (here $\boldsymbol{c} = Q^{-1}\boldsymbol{b}$), we still have a multivariate normal distribution.

We could even reach variances $= 1$ by including $\sqrt{\Lambda}$ in the change from $\boldsymbol{b}$ to $\boldsymbol{z}$:

Standard normal $\quad \boldsymbol{b} = \sqrt{\Lambda}Q\boldsymbol{z}$ changes $p(\boldsymbol{b})d\boldsymbol{b}$ to $p(\boldsymbol{z})d\boldsymbol{z} = \dfrac{e^{-\boldsymbol{z}^{\mathrm{T}}\boldsymbol{z}/2}}{(2\pi)^{m/2}}d\boldsymbol{z}$

This tells us the right weight matrix W to bring $A\boldsymbol{x} = \boldsymbol{b}$ back to ordinary least squares for $WA\boldsymbol{x} = W\boldsymbol{b}$. We want $W\boldsymbol{b}$ to become the standard normal $\boldsymbol{z}$. So W will be the inverse of $\sqrt{\Lambda}Q$. Better than that, $\boldsymbol{C = W^{\mathrm{T}}W}$ ***is the inverse of*** $\boldsymbol{Q\Lambda Q^{\mathrm{T}}}$ ***which is*** $\boldsymbol{\Sigma}$.

Summary For independent errors, Σ is the diagonal matrix $\mathbf{diag}(\sigma_1^2, \ldots, \sigma_m^2)$. This is the usual choice. The right weights w_i for the equations $A\boldsymbol{x} = \boldsymbol{b}$ are $1/\sigma_1, \ldots, 1/\sigma_m$ (this will equalize all variances to 1). The right matrix $C = W^{\mathrm{T}}W$ in the middle of the weighted least squares equations is exactly Σ^{-1}:

Weighted least squares $\qquad A^{\mathrm{T}}\Sigma^{-1}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\Sigma^{-1}\boldsymbol{b}$ $\qquad$ (9)

This choice of weighting returns $A\boldsymbol{x} = \boldsymbol{b}$ to a least squares problem $WA\boldsymbol{x} = W\boldsymbol{b}$ with equally reliable and independent errors. The usual equation $(WA)^{\mathrm{T}}WA\widehat{\boldsymbol{x}} = (WA)^{\mathrm{T}}W\boldsymbol{b}$ is the same as (9).

It was Gauss who found this *best linear unbiased estimate* $\widehat{\boldsymbol{x}}$. Unbiased because the mean of $\boldsymbol{x} - \widehat{\boldsymbol{x}}$ is zero, linear because of equation (9), best because the covariance of $\boldsymbol{x} - \widehat{\boldsymbol{x}}$ is as small as possible. That covariance (for error in $\widehat{\boldsymbol{x}}$, not error in $\boldsymbol{b}$!) is important:

Covariance of the best $\widehat{\boldsymbol{x}}$ $\qquad P = \mathrm{E}\left[(\boldsymbol{x} - \widehat{\boldsymbol{x}})(\boldsymbol{x} - \widehat{\boldsymbol{x}})^{\mathrm{T}}\right] = \left(A^{\mathrm{T}}\Sigma^{-1}A\right)^{-1}.$ $\qquad$ (10)

Example 5 Your pulse rate is measured ten times by independent doctors, all equally reliable. The mean error of each b_i is zero, and each variance is σ^2. Then $\Sigma = \sigma^2 I$. The ten equations $x = b_i$ produce the 10 by 1 matrix A of all ones. The best estimate $\widehat{x}$ is the average of the ten b_i. *The variance of that average value* $\widehat{x}$ *is the number* P:

$$P = (A^{\mathrm{T}}\Sigma^{-1}A)^{-1} = \sigma^2/10 \quad \text{so averaging reduces the variance.}$$

This matrix $P = (A^{\mathrm{T}}\Sigma^{-1}A)^{-1}$ tells how reliable is the result $\widehat{\boldsymbol{x}}$ of the experiment (Problem 6). P does not depend on the b's in the actual experiment! Those b's have probability distributions. Each experiment produces a sample value of $\widehat{\boldsymbol{x}}$ from a sample $\boldsymbol{b}$.

When a small Σ gives good reliability of the inputs $\boldsymbol{b}$, a small P gives good reliability of the outputs $\widehat{x}$. The key formula $P = (A^{\mathrm{T}}\Sigma^{-1}A)^{-1}$ connects those covariances.

Principal Component Analysis

These paragraphs are about finding useful information in a data matrix A. Start by measuring m properties (m features) of n samples. These could be grades in m courses for n students (a row for each course, a column for each student). From each row, subtract its average so the sample means are zero. We look for a ***combination of courses*** and/or ***combination of students*** for which the data provides the most information.

Information is "distance from randomness" and it is measured by **variance**. A large variance in course grades means greater information than a small variance.

The key matrix idea is the Singular Value Decomposition $A = U\Sigma V^{\mathrm{T}}$. We are back again to $A^{\mathrm{T}}A$ and AA^{T}, because their unit eigenvectors are the singular vectors $v_1, \ldots, v_n$ in V and $u_1, \ldots, u_m$ in U. The singular values in the diagonal matrix Σ (not the covariance) are in decreasing order and σ_1 is the most important. Weighting the m courses by the components of u_1 gives a "***master course***" or "***eigencourse***" with the most significant grades.

Example 6 Suppose the grades A, B, C, F are worth $\mathbf{4, 2, 0, -6}$ points. If each course and each student has one of each grade, then all means are zero. Here is the grade matrix A with $(1, 1, 1, 1)$ in its nullspace (rank 3). To keep integers, the SVD of A will be written as $2U$ times $\Sigma/4$ times $(2V)^{\mathrm{T}}$. So the σ's are $12, 8, 4$:

$$\begin{bmatrix} -6 & 2 & 0 & 4 \\ 0 & 4 & -6 & 2 \\ 4 & 0 & 2 & -6 \\ 2 & -6 & 4 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 1 & -1 \\ -1 & -1 & 1 \\ 1 & -1 & -1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & & \\ & 2 & \\ & & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 & -1 \\ -1 & -1 & 1 & 1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

Weighting the rows (the courses) by $u_1 = \frac{1}{2}(-1, -1, 1, 1)$ will give the *eigencourse*. Weighting the columns (the students) by $v_1 = \frac{1}{2}(1, -1, 1, -1)$ gives the *eigenstudent*. The fraction of the grade matrix that is "explained" by that one course and student is $\sigma_1^2/(\sigma_1^2 + \sigma_2^2 + \sigma_3^2) = 9/14$. The σ's in the SVD are the variances σ^2.

I guess this master course is what a Director of Admissions is looking for. If all grades in gym are the same, that row of A will be all zero—and gym is not part of the master course. Probably calculus is a part, but what about students who don't take calculus? The problem of ***missing data*** (holes in the matrix A) is extremely difficult for social sciences and the census and so much of the statistics of experiments.

Gene expression data Determining the functions of genes, and combinations of genes, is a central problem of genetics. Which genes combine to give which properties? Which genes malfunction to give which diseases?

We now have an incredibly fast way to find gene expression data in the lab. A gene microarray is often packed onto an Affymetrix chip, measuring tens of thousands of genes from one sample (one person). The understanding of genetic data (*bioinformatics*) has become a tremendous application of linear algebra.

Problem Set 8.6

1 Which line $Ct + D$ is the best fit to the three independent measurements $1, 2, 4$ at times $t = 0, 1, 2$ if the variances $\sigma_1^2, \sigma_2^2, \sigma_3^2$ are $1, 1, 2$? Use weights $w_i = 1/\sigma_i$.

2 In Problem 1, suppose that the third measurement is **totally unreliable**. The variance σ_3^2 becomes infinite. Then the best line will not use _____. Find the line that goes through the first two points and solves the first two equations in $Ax = b$ exactly.

3 In Problem 1, suppose that the third measurement is **totally reliable**. The variance σ_3^2 approaches zero. Now the best line will go through the third point exactly. Choose that line to minimize the sum of squares of the first two errors.

4 A single flip of a fair coin (0 or 1) has mean $m = 1/2$ and variance $\sigma^2 = 1/4$. This was Example 1. For the sum of two flips, the mean is $m = 1$. Compute the variance σ^2 around this mean, using the outcomes $0, 1, 2$ with their probabilities.

5 Instead of adding the flip results, make them two independent experiments. The outcome is $(0, 0), (1, 0), (0, 1)$ or $(1, 1)$. What is the covariance matrix Σ?

6 Change Example 1 so that the coin flip can be unfair. The probability is p for heads and $1 - p$ for tails. Find the mean m and the variance σ^2 of this distribution.

7 For two independent measurements $x = b_1$ and $x = b_2$, the best $\widehat{x}$ should be some weighted average $\widehat{x} = ab_1 + (1 - a)b_2$. When b_1 and b_2 have mean zero and variances σ_1^2 and σ_2^2, the variance of $\widehat{x}$ will be $P = a^2\sigma_1^2 + (1-a)^2\sigma_2^2$. ***Choose the number a that minimizes P***: $dP/da = 0$.

Show that this a gives the $\widehat{x}$ in equation (2) which the text claimed is best, using weights $w_1 = 1/\sigma_1$ and $w_2 = 1/\sigma_2$.

8 The least squares estimate correctly weighted by Σ^{-1} is $\widehat{x} = (A^{\mathrm{T}}\Sigma^{-1}A)^{-1}A^{\mathrm{T}}\Sigma^{-1}b$. ***Call that*** $\widehat{x} = Lb$. If b contains an error vector e, then $\widehat{x}$ contains the error Le.

The covariance matrix of those output errors Le is their expected value (average value) $P = \mathrm{E}\left[(Le)(Le)^{\mathrm{T}}\right] = L\mathrm{E}\left[ee^{\mathrm{T}}\right]L^{\mathrm{T}} = L\Sigma L^{\mathrm{T}}$. **Problem**: Do the multiplication $L\Sigma L^{\mathrm{T}}$ to show that P equals $(A^{\mathrm{T}}\Sigma^{-1}A)^{-1}$ as predicted in equation (10).

9 Change the grades to $3, 1, -1, -3$ for A, B, C, F. Show that the SVD of this grade matrix has the same u_1, u_2, v_1, v_2 (same eigencourses) as in Example 5, but now A has rank 2.

Grade matrix
$$A = \begin{bmatrix} 3 & -1 & 1 & -3 \\ -1 & 3 & -3 & 1 \\ -3 & 1 & -1 & 3 \\ 1 & -3 & 3 & -1 \end{bmatrix}$$

Notes One way to deal with missing entries in A is to complete the matrix to have minimum rank. And statistics makes major use of the pseudoinverse A^+ (which is exactly the left inverse $(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ from the normal equation when $A^{\mathrm{T}}A$ is invertible).